

Jake Burttram

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EDUCATION

Clemson University

Bachelor of Science in Mechanical Engineering

Palmetto Fellows Scholar

Clemson, SC

May 2024

3.61 GPA

PROFESSIONAL EXPERIENCE

Raycap

Mechanical Engineer I

North Charleston, SC

July 2024-Present

- Own mechanical designs from customer and site requirements through 3D models, fabrication drawings, BOMs, and contractor installation documentation.
- Create parametric designs in Autodesk Inventor with an emphasis on design for manufacturing, assembly, installation, and efficient reuse of proven designs.
- Design steel and composite assemblies around in-house fabrication capabilities including cutting, forming, welding, galvanizing, and composite fabrication.
- Develop structural concepts and coordinate with structural engineers for analysis, refinement, and approval.

Electrolux (Co-op)

Research and Development Co-op

Anderson, SC

January 2022-August 2023

- Managed four cost-saving projects generating over \$545,000 in annual savings.
- Managed, documented, and presented results from over 30 concurrent quality tests for a new refrigerator model.
- Led manufacturing trials to qualify new components for production and supported product development through CAD updates, drawings, and tolerance analysis.
- Initiated and tested a refrigerator noise-reduction concept and analyzed service-call data to identify and prioritize product quality issues.

SKILLS AND CREDENTIALS

Skills

- CAD & Design (Autodesk Inventor, AutoCAD, SolidWorks, Fusion 360, CATIA V5, Siemens NX, Parametric Modeling)
- Design for Manufacturing (Sheet Metal & Weldment Design, CNC/Laser-Cut Parts, Composite Fabrication, Tolerance Analysis, 3D Printing)
- Engineering Tools (Autodesk Vault, Autodesk CFD, COMSOL Multiphysics, Microsoft Excel, MATLAB, Python)
- Product Development (Engineering Change Management, DFMEA/PFMEA, PPAP, Prototyping, Test Planning & Documentation)

Professional Credential: Passed FE Mechanical Exam, 2024

Certifications: CSWA 2019, SolidWorks 2020